Fiscal Year:	FY 2004	**Task Last Updated:**	FY 10/20/2004
PI Name:	Marston, Philip L. Ph.D.		
Project Title:	Passive and Active Stabilization of Liquid Bridges in Low Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marston@wsu.edu	**Fax:**	FY 509-335-7816
PI Organization Type:	UNIVERSITY	**Phone:**	509-335-5343
Organization Name:	Washington State University		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:			
City:	Pullman	**State:**	WA
Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/10/2000	**End Date:**	03/09/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Niederhaus, Charles	**Contact Phone:**	(216) 433-5461
Contact Email:	charles.niederhaus@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thiessen, David (Washington State University)		
Grant/Contract No.:	NAG3-2378		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	There are two general types of stabilization methods being investigated. These use feedback control of radial electrostatic stresses or the acoustic radiation pressure of high frequency sound waves. In the electrostatic approach the spatial distribution of the Maxwell stress on the bridge is actively adjusted so as to counteract the growth of the most unstable mode. This is achieved by optically sensing the shape of the bridge and electronic control of the voltage applied to an array of ring-shaped electrodes that are centered on the bridge. Acoustic stabilization of liquid bridges is investigated with appropriately designed ultrasonic fields. One approach involves the selection of the acoustic field such that a local variation in the bridge radius automatically produces a compensating change in the local acoustic radiation pressure. In another approach the spatial distribution of the acoustic radiation pressure on the bridge is actively adjusted so as to counteract the growth of the most unstable mode. This is achieved by optically sensing the shape of the bridge as in the electrostatic case. Active reduction of the response of liquid bridges to g-jitter should also be possible with an appropriate feedback algorithm. The bridges for investigation are primarily liquid bridges in air, however some bridges		

	surrounded by an immiscible host liquid are also studied. The primary objective is to stabilize liquid bridges against break up from the Rayleigh-Plateau instability. A secondary objective is to reduce the response of liquid bridges to g-jitter using active damping and modification of the natural frequency of bridge oscillations. The research should give improved capabilities to manage fluids in low-gravity by achieving stabilization beyond the Rayleigh-Plateau limit and by reducing the response of liquid bridges to g-jitter. This research may have application to float-zone crystal growth processes or to the containerless management of liquids and the production and control of liquid drops. Long liquid bridges should be beneficial to float zone processing because it is possible to reduce thermal gradients in the liquid as well as in the solid. Reducing thermal stresses in newly formed crystals can be important to the crystal quality. The ability to provide active damping may have significant benefits for environments having significant g-jitter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The stabilization and active damping methods developed ate not limited in applicability to liquid bridges in reduced gravity since they may be scaled to earth-based situations involving free liquid interfaces.
Task Progress:	The stability of liquids having a free surface in reduced gravity is affected by capillary forces. Liquids having free surfaces can display also large responses to ambient vibrations. The research supported by this grant concerns generic approaches to the stabilization of liquid systems and to suppressing their response to ambient vibration. The specific system selected for investigation is a liquid bridge either in air or in a Plateau tank of immiscible density-matched liquids. The primary accomplishments during the grant year under consideration are as follows. (1) The primary effort is an experimental investigation and an improved theory for the active damping of the low frequency capillary modes of liquid bridges. This control method is based on optically sensing the mode amplitude and mode velocity and using that information to actively control the electrostatic stresses applied to the bridge by an array of electrodes. This is done in such a way that in active damping the mechanical energy is converted to electrical energy and is dissipated. (2) In other research supported by this grant we have analyzed the acoustic radiation force on fluid columns in an acoustic standing wave. This was important since we have previously demonstrated that radiation pressure can be used to be able to stabilize liquid columns. Our recent publication on the radiation force [W. Wei, D. B. Thiessen, and P. L. Marston, J. Acoust. Soc. Am. 116, 201-208 (2004)] considers liquid columns in air, liquid columns in a Plateau tank, flames in air, and bubbles in water. Prior work supported by this grant emphasized reduced-gravity experiments on liquid bridges in air carried out on NASA's KC-135 aircraft in the following areas: (1) passive stabilization of liquid bridges in air using an ultrasonic standing wave [Marr-Lyon et al., Phys. Rev. Lett. 86, 2293-2296 (2001) & 87, art. no. 209901 (2001)] and (2) active stabilization of liquid bridges in air based on the active control of electrostatic stresses on the bridge [Thiessen et al. Journal of Fluid Mechanics 457, 285-294 (2002)]. With both of these methods we found that it is possible to stabilize a cylindrical bridge far beyond the natural Rayleigh-Plateau limiting length L given by the circumference of the liquid bridge. Experience working on the KC-135 indicates that for the purpose of understanding the active damping of capillary modes it would be highly desirable to identify a liquid-liquid combination for the use in a Plateau tank where the intrinsic damping due to viscosity would be small. This is because the technical demands of the KC-135 made it desirable to study active damping in a table-top experiment. We identified a suitable combination of liquids and have been using data from that system to test models of the dynamics of actively damped liquid bridges [Thiessen et al. accepted for publication]. Even for the combination of liquids used however, it has been important to improve our understanding of viscous contributions to the damping and we extended or theoretical and computational models of that damping. The active electrostatic damping of capillary modes is a new technology that advances the state-of-art stabilizing capillary systems. We also extended our understanding of stabilizing of liquid bridges based on acoustic radiation pressure.
Bibliography Type:	Description: (Last Updated: 06/21/2021)
Abstracts for Journals and Proceedings	Thiessen, D. B., Wei, W., and Marston, P. L. "Ultrasonic stabilization of small lifted flames" J. Acoust. Soc. Am. 114, 2388 (2003) J. Acoust. Soc. Am. 114, 2388 (2003) , Nov-2003
Abstracts for Journals and Proceedings	Marston, P. L. "Shape oscillations of acoustically levitated drops in water: Early research with Bob Apfel on modulated radiation pressure" (invited) J. Acoust. Soc. Am. 115, 2419 (2004) J. Acoust. Soc. Am. 115, 2419 (2004) , May-2004
Abstracts for Journals and Proceedings	Wei, W., Thiessen, D. B., and Marston, P. L. "Radiation force on a compressible cylinder in a standing wave: Long-wavelength approximation" J. Acoust. Soc. Am. 114, 2356 (2003) J. Acoust. Soc. Am. 114, 2356 (2003) , Nov-2003
Articles in Peer-reviewed Journals	Wei, W., Thiessen, D. B., and Marston, P. L. "Acoustic radiation force on a compressible cylinder in a standing wave" Journal of the Acoustical Society of America 116, 201-208 (2004) , Jul-2004
Presentation	Wei, W., Thiessen, D. B., and Marston, P. L. "Enhanced damping for capillary bridge oscillation using velocity feedback (abstract)" Pacific Northwest Meeting of the American Physical Society May-2004
Significant Media Coverage	NA "Announcement of Acoustical Society of America Silver Medal in Physical Acoustics (2003) to Marston “...for contributions to generalized ray theories for acoustical scattering and the acoustical manipulation of fluids to study fundamental phenomena in fluid mechanics and optics.“" Journal of the Acoustical Society of America (Supplement) & ASA Press release Vol. 114, No. 4, Part 2 , Oct-2003